Search in a Sorted Infinite Array (medium)

Problem Statement

Given an infinite sorted array (or an array with unknown size), find if a given number 'key' is present in the array. Write a function to return the index of the 'key' if it is present in the array, otherwise return -1.

Since it is not possible to define an array with infinite (unknown) size, you will be provided with an interface `ArrayReader` to read elements of the array. `ArrayReader.get(index)` will return the number at index; if the array's size is smaller than the index, it will return `Integer.MAX_VALUE`.

Example 1:

```
Input: [4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30], key = 16
Output: 6
Explanation: The key is present at index '6' in the array.
```

Example 2:

```
Input: [4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30], key = 11
Output: -1
Explanation: The key is not present in the array.
```

Example 3:

```
Input: [1, 3, 8, 10, 15], key = 15
Output: 4
Explanation: The key is present at index '4' in the array.
```

Example 4:

```
Input: [1, 3, 8, 10, 15], key = 200
Output: -1
Explanation: The key is not present in the array.
```

Try it yourself

Try solving this question here:

Java

```
/*class ArrayReader {
  int[] arr;

  ArrayReader(int[] arr) {
    this.arr = arr;
  }

  public int get(int index) {
    if (index >= arr.length)
      return Integer.MAX_VALUE;
    return arr[index];
  }

}*/


class Solution {

  public static int searchInfiniteSortedArray(ArrayReader reader, int key) {
    // TODO: Write your code here
    return -1;
  }
}
```

Test Case | Run Code Result | Stdout

```
[4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30]
16
```

Use Example Testcases | Run | Submit

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Solution: Number Range

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Solution: Search In A Sorted Infinite Array

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